Задания. Раздел 1.

Лог работы – команды и результаты в записать в файл practice2_${MYUSER}_part1.log, где MYUSER – переменная с именем пользователя практики 1.

1. Создайте на сервере для 1 практики ключ ssh при помощи программы ssh-keygen
2. Скопируйте созданный ключ на сервер для 2 практики для пользователя root при помощи программы ssh-copy-id
3. Подключитесь к серверу для 2 практики под пользователем root и сравните содержимое файла открытого ключа на сервере 1 ~/.ssh/*.pub и файла ~/.ssh/authorized_keys на сервере для 2 практики, а так же права доступа для каждого из файлов

Все последующие пункты – для сервера 2 практики

4. Создайте пользователя user1 при помощи команды useradd, укажите необходимость создания домашнего каталога и shell /bin/bash. Создайте пароль пользователю user1
5. Создайте пользователя user2 и user3 при помощи команды adduser
6. Для пользователя user3 смените shell на /usr/sbin/nologin (man usermod), выполните вход под этим пользователем при помощи утилиты su, сначала без дополнительных параметров, затем с явным указанием shell /bin/bash в параметрах su. Выполните logout
7. Создайте новую группу и добавьте её для всех пользователей user* как дополнительную, посмотрите список групп всех пользователей user*

```
eltex-pg2-v8@eltex-2026-spring:~$ touch practice2_`whoami`_part1.log
eltex-pg2-v8@eltex-2026-spring:~$ ls
2026-01  2026-08                        eltex-pg2-v8_part3.log
2026-02  2026-09                        eltex-pg2-v8_part.log
2026-03  2026-10                        _part.log
2026-04  2026-11                        practice2_eltex-pg2-v8_part1.log
2026-05  2026-12                        report_part1.txt
2026-06  D1                             screenlog.0
2026-07  eltex-pg2-v8_part3_gt50M.log
```

1. ssh-keygen

```
eltex-pg2-v8@eltex-2026-spring:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/eltex-pg2-v8/.ssh/id_rsa):
Created directory '/home/eltex-pg2-v8/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/eltex-pg2-v8/.ssh/id_rsa
Your public key has been saved in /home/eltex-pg2-v8/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:8lhcXdX3ramGxwOtpvRHtVOTYS4q/Z0wLditSdu9iAU eltex-pg2-v8@eltex-2026-sprin
g
The key's randomart image is:
+---[RSA 3072]----+
|              ..o|
|           . . oo|
|          . . o *|
|       . .   ..++|
|      . S .E+.+=.|
|       = ..++*=o |
|      . o .*ooX.o|
|       . .+ X=.+.|
|        .o.= o ..|
+----[SHA256]-----+
eltex-pg2-v8@eltex-2026-spring:~$
```

2. ssh-copy-id -i ~/.ssh/id_rsa.pub root@172.16.9.208

```
eltex-pg2-v8@eltex-2026-spring:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub root@172.16.9.208
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/eltex-pg2-v8/.ssh/id_rsa.pub"
The authenticity of host '172.16.9.208 (172.16.9.208)' can't be established.
ED25519 key fingerprint is SHA256:YNVzj+RtetZfXv5XUn/CUjeEcaMD4eg3jy8EuGGIS+o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@172.16.9.208's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@172.16.9.208'"
and check to make sure that only the key(s) you wanted were added.
```

3.

С первого сервера

```
eltex-pg2-v8@eltex-2026-spring:~$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDSsnKEOJYUJz9MapV346rKBX3MtItXmM2RN0XokPdD
PVf8VChXwvI5r9OMt4KMPJarBBZoo1ECAG+xfZJBcR0ZbaRqDJ6wcSZ5f8ZV/IvUlhG/YqEFEf/rwf6M
xINcYF1plTzcF2qjJ4y1Fd3nZRflppKlUQBzykVdGxYlhC3n8RMJhxTaiExXjQlXXmc308p9zQ/ZLR0L
RFevozTjQfpFZ9cXdT0Fit1wY184VnULkUIqFhVtZBEh3ddsMPaazX8KbmnRtaXeRRSZ02XZbpakglDa
jpQbkrYgqkVw92Cd43937tz6ow/tqRMqjfcT9dnUTHUsLa/Dbtgt4XKM6BYcT1HdTcgWGD9vQdYcMYbl
9U3AUMSUxqwcqyy81hUEXEFQPNPqZsyEBgVIjvCirWFR6HqpB5LgOlnvEN0KhRtz+yCePwhQC9TWBmZc
7leo+pEQjcETZftKW2U+bh7SDCwaksa4DD5eUDOK/NX6OI+z9OMw7cujqOmz2zyfI9IKhV0= eltex-p
g2-v8@eltex-2026-spring
eltex-pg2-v8@eltex-2026-spring:~$ ls -ld ~/.ssh/id_rsa.pub
-rw-r--r-- 1 eltex-pg2-v8 eltex-pg2-v8 584 Mar 12 17:50 /home/eltex-pg2-v8/.ssh/
id_rsa.pub
eltex-pg2-v8@eltex-2026-spring:~$
```

С второго сервера

```
root@eltex-practice2-pg2-v8:~# cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDSsnKEOJYUJz9MapV346rKBX3MtItXmM2RN0XokPdD
PVf8VChXwvI5r9OMt4KMPJarBBZoo1ECAG+xfZJBcR0ZbaRqDJ6wcSZ5f8ZV/IvUlhG/YqEFEf/rwf6M
xINcYF1plTzcF2qjJ4y1Fd3nZRflppKlUQBzykVdGxYlhC3n8RMJhxTaiExXjQlXXmc308p9zQ/ZLR0L
RFevozTjQfpFZ9cXdT0Fit1wY184VnULkUIqFhVtZBEh3ddsMPaazX8KbmnRtaXeRRSZ02XZbpakglDa
jpQbkrYgqkVw92Cd43937tz6ow/tqRMqjfcT9dnUTHUsLa/Dbtgt4XKM6BYcT1HdTcgWGD9vQdYcMYbl
9U3AUMSUxqwcqyy81hUEXEFQPNPqZsyEBgVIjvCirWFR6HqpB5LgOlnvEN0KhRtz+yCePwhQC9TWBmZc
7leo+pEQjcETZftKW2U+bh7SDCwaksa4DD5eUDOK/NX6OI+z9OMw7cujqOmz2zyfI9IKhV0= eltex-p
g2-v8@eltex-2026-spring
root@eltex-practice2-pg2-v8:~# ls -ld ~/.ssh/authorized_keys
-rw------- 1 root root 584 Mar 12 10:55 /root/.ssh/authorized_keys
root@eltex-practice2-pg2-v8:~#
```

4. useradd -m -s /bin/bash user1 ; passwd user1

```
root@eltex-practice2-pg2-v8:~# useradd -m -s /bin/bash user1
useradd: user 'user1' already exists
root@eltex-practice2-pg2-v8:~# passwd user1
New password:
Retype new password:
passwd: password updated successfully
root@eltex-practice2-pg2-v8:~#
```

5. adduser user2; adduser user3

```
root@eltex-practice2-pg2-v8:~# adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1002) ...
info: Adding new user `user2' (1002) with group `user2 (1002)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `user2' to supplemental / extra groups `users' ...
info: Adding user `user2' to group `users' ...
```

```
root@eltex-practice2-pg2-v8:~# adduser user3
info: Adding user `user3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user3' (1003) ...
info: Adding new user `user3' (1003) with group `user3 (1003)' ...
info: Creating home directory `/home/user3' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user3
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `user3' to supplemental / extra groups `users' ...
info: Adding user `user3' to group `users' ...
```

6. usermod -s /usr/sbin/nologin user 3 ; su user3; su -s /bin/bash user3 ; exit

```
root@eltex-practice2-pg2-v8:~# usermod -s /usr/sbin/nologin user3
root@eltex-practice2-pg2-v8:~# su user3
This account is currently not available.
root@eltex-practice2-pg2-v8:~# su -s /bin/bash user3
user3@eltex-practice2-pg2-v8:/root$ logout
bash: logout: not login shell: use `exit'
user3@eltex-practice2-pg2-v8:/root$ exit
exit
root@eltex-practice2-pg2-v8:~# Z
```

7. groupadd users_group ; usermod -aG users_group user1 ; usermod -aG users_group user2 ; usermod -aG users_group user3 ; groups user1 ; groups user2 ; groups user3

```
root@eltex-practice2-pg2-v8:~# groupadd users_group
root@eltex-practice2-pg2-v8:~# usermod -aG users_group user1
root@eltex-practice2-pg2-v8:~# usermod -aG users_group user2
root@eltex-practice2-pg2-v8:~# usermod -aG users_group user3
```

```
root@eltex-practice2-pg2-v8:~# groups user1
user1 : groups: cannot find name for group ID 1001
1001 users_group
root@eltex-practice2-pg2-v8:~# groups user2
user2 : user2 users users_group
root@eltex-practice2-pg2-v8:~# groups user3
user3 : user3 users users_group
```

Задания. Раздел 1.

9. Создайте каталог /opt/share и назначьте группу из предыдущего пункта его владельцем, установите на этот каталог бит SGID, права для группы rwx.
10. Для user1 задайте перманентно umask, снимающий право чтения для «прочих»
11. Создайте каждым из пользователей новые файлы в каталоге /opt/share, удалите файлы созданные другими пользователями
12. Повторите предыдущий пункт, предварительно установив sticky bit на каталоге /opt/share
13. Разрешите user1 выполнять привилегированную команду dmesg при помощи команды sudo, а user2 – при помощи скрипта на языке bash с установленным флагом SUID
14. Для всех пользователей user* задайте время действия пароля – 10 дней.
15. Отредактируйте файл /etc/motd, вписав туда свое имя и фамилию
16. Создайте копию содержимого каталога /etc в каталог /root/etc_backup при помощи программы rcync
17. Заархивируйте содержимое каталога /root/etc_backup архиватором tar, используйте алгоритмы сжатия gzip, bzip2, 7zip, сравните размеры полученных файлов
18. Отредактируйте файл /etc/motd, вписав туда текущую дату и время, синхронизируйте каталог /root/etc_backup с каталогом /etc при помощи rsync, добавьте файл motd в архив, сжатый gzip
19. Сравните содержимое архива, упакованного bzip2 с сорежимым каталога /root/etc_backup
20. Распакуйте архивы etc_backup, упакованные gzip и 7zip в каталоги /root/etc_backup_gzip и /root/etc_backup_7zip, сравните программой diff файлы motd в этих каталогах.

---

9. mkdir -p /opt/share ; chown :users_group /opt/share ; chmod 2770 /opt/share ; ls -ld /opt/share

```
root@eltex-practice2-pg2-v8:~# mkdir -p /opt/share
root@eltex-practice2-pg2-v8:~# chown :users_group /opt/share
root@eltex-practice2-pg2-v8:~# chmod 2770 /opt/share
root@eltex-practice2-pg2-v8:~# ls -ld /opt/share
drwxrws--- 2 root users_group 4096 Mar 13 07:37 /opt/share
root@eltex-practice2-pg2-v8:~#
```

10. echo "umask 027" >> /home/user1/.bashrc ; su - user1 ; umask ; exit

```
root@eltex-practice2-pg2-v8:~# echo "umask 027" >> /home/user1/.bashrc
root@eltex-practice2-pg2-v8:~# su - user1
groups: cannot find name for group ID 1001
user1@eltex-practice2-pg2-v8:~$ umask
0027
user1@eltex-practice2-pg2-v8:~$ exit
logout
root@eltex-practice2-pg2-v8:~#
```

11. su - user1 -c "touch /opt/share/file_user1" ; su - user2 -c "touch /opt/share/file_user2" ; su -s /bin/bash - user3 -c "touch /opt/share/file_user3" ; ls -l /opt/share ; su - user1 -c "rm /opt/share/file_user2" ; su - user2 -c "rm /opt/share/file_user3" ; ls -l /opt/share

```
root@eltex-practice2-pg2-v8:~# su - user1 -c "touch /opt/share/file_user1"
root@eltex-practice2-pg2-v8:~# su - user2 -c "touch /opt/share/file_user2"
root@eltex-practice2-pg2-v8:~# su -s /bin/bash - user3 -c "touch /opt/share/file_user3"
```

```
root@eltex-practice2-pg2-v8:~# ls -l /opt/share
total 0
-rw-r--r-- 1 user1 users_group 0 Mar 13 08:06 file_user1
-rw-rw-r-- 1 user2 users_group 0 Mar 13 08:06 file_user2
-rw-rw-r-- 1 user3 users_group 0 Mar 13 08:06 file_user3
root@eltex-practice2-pg2-v8:~#
```

```
root@eltex-practice2-pg2-v8:~# su - user1 -c "rm /opt/share/file_user2"
root@eltex-practice2-pg2-v8:~# su - user2 -c "rm /opt/share/file_user3"
root@eltex-practice2-pg2-v8:~# ls -l /opt/share
total 0
-rw-r--r-- 1 user1 users_group 0 Mar 13 08:06 file_user1
root@eltex-practice2-pg2-v8:~#
```

12. chmod +t /opt/share ; su - user1 -c "touch /opt/share/file_user1" ; su - user2 -c "touch /opt/share/file_user2" ; su -s /bin/bash - user3 -c "touch /opt/share/file_user3" ; ls -ld /opt/share ; su - user1 -c "rm /opt/share/file_user2" ; su - user2 -c "rm /opt/share/file_user3" ;

```
root@eltex-practice2-pg2-v8:~# chmod +t /opt/share
root@eltex-practice2-pg2-v8:~# su - user1 -c "touch /opt/share/file_user1"
root@eltex-practice2-pg2-v8:~# su - user2 -c "touch /opt/share/file_user2"
root@eltex-practice2-pg2-v8:~# su -s /bin/bash - user3 -c "touch /opt/share/file_user3"
root@eltex-practice2-pg2-v8:~# ls -ld /opt/share
drwxrws--T 2 root users_group 4096 Mar 13 08:12 /opt/share
```

```
root@eltex-practice2-pg2-v8:~# su - user1 -c "rm /opt/share/file_user2"
rm: cannot remove '/opt/share/file_user2': Operation not permitted
root@eltex-practice2-pg2-v8:~# su - user2 -c "rm /opt/share/file_user3"
rm: cannot remove '/opt/share/file_user3': Operation not permitted
```

13. visudo -f /etc/sudoers.d/user1_dmesg ; su - user1 -c "sudo dmesg | head -5" ; vim /usr/local/bin/dmesg_suid.sh; chown root:root /usr/local/bin/dmesg_suid.sh ; chmod u+s

/usr/local/bin/dmesg_suid.sh ; su - user2 -c "/usr/local/bin/dmesg_suid.sh | head -5" ;

```
root@eltex-practice2-pg2-v8:~# visudo -f /etc/sudoers.d/user1_dmesg
```

```
  GNU nano 7.2                    /etc/sudoers.d/user1_dmesg.tmp
user1 ALL=(ALL) NOPASSWD: /bin/dmesg

                                 [ Read 1 line ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

```
root@eltex-practice2-pg2-v8:~# su - user1 -c "sudo dmesg | head -5"
[    0.000000] Linux version 6.8.0-55-generic (buildd@lcy02-amd64-095) (x86_64-linux-gnu-gcc-13 (Ubu
ntu 13.3.0-6ubuntu2~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #57-Ubuntu SMP PREEMPT_DYN
AMIC Wed Feb 12 23:42:21 UTC 2025 (Ubuntu 6.8.0-55.57-generic 6.8.12)
[    0.000000] Command line: BOOT_IMAGE=/vmlinuz-6.8.0-55-generic root=/dev/mapper/ubuntu--vg-ubuntu
--lv ro
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
```

```
root@eltex-practice2-pg2-v8:~# vim /usr/local/bin/dmesg_suid.sh
```

```
#!/bin/bash
/bin/dmesg
EOF
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
:wq
```

```
root@eltex-practice2-pg2-v8:~# chown root:root /usr/local/bin/dmesg_suid.sh
root@eltex-practice2-pg2-v8:~# chmod u+s /usr/local/bin/dmesg_suid.sh
```

```
root@eltex-practice2-pg2-v8:~# su - user2 -c "/usr/local/bin/dmesg_suid.sh | head -5"
-bash: line 1: /usr/local/bin/dmesg_suid.sh: Permission denied
```

14. chage -M 10 "user1" ; chage -M 10 "user2" ; chage -M 10 "user3" ; chage -l user1 | grep Maximum; chage -l user2 | grep Maximum ; chage -l user3 | grep Maximum

```
root@eltex-practice2-pg2-v8:~# chage -M 10 "user1"
root@eltex-practice2-pg2-v8:~# chage -M 10 "user2"
root@eltex-practice2-pg2-v8:~# chage -M 10 "user3"
root@eltex-practice2-pg2-v8:~# chage -l user1 | grep Maximum
Maximum number of days between password change          : 10
root@eltex-practice2-pg2-v8:~# chage -l user2 | grep Maximum
Maximum number of days between password change          : 10
root@eltex-practice2-pg2-v8:~# chage -l user3 | grep Maximum
Maximum number of days between password change          : 10
```

15. echo "Плотницкий Максим" >> /etc/motd ; cat /etc/motd

```
root@eltex-practice2-pg2-v8:~# echo "Плотницкий Максим" >> /etc/motd
root@eltex-practice2-pg2-v8:~# cat /etc/motd
Плотницкий Максим
```

16. rsync -av /etc/ /root/etc_backup/

```
root@eltex-practice2-pg2-v8:~# rsync -av /etc/ /root/etc_backup/
sending incremental file list
created directory /root/etc_backup
./
.pwd.lock
.resolv.conf.systemd-resolved.bak
.updated
adduser.conf
bash.bashrc
bash_completion
bindresvport.blacklist
ca-certificates.conf
crontab
crypttab
debconf.conf
debian_version
deluser.conf
```

```
sent 2,451,278 bytes  received 19,410 bytes  197,655.04 bytes/sec
total size is 2,366,800  speedup is 0.96
```

17. apt install -y bzip2 p7zip-full ; tar czf /root/etc_backup.tar.gz -C /root etc_backup ; tar cjf /root/etc_backup.tar.bz2 -C /root etc_backup ; 7z a /root/etc_backup.7z /root/etc_backup ; ls -lh /root/etc_backup.*

```
root@eltex-practice2-pg2-v8:~# tar czf /root/etc_backup.tar.gz -C /root etc_backup
root@eltex-practice2-pg2-v8:~# tar cjf /root/etc_backup.tar.bz2 -C /root etc_backup
root@eltex-practice2-pg2-v8:~# 7z a /root/etc_backup.7z /root/etc_backup

7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
 64-bit locale=en_US.UTF-8 Threads:2 OPEN_MAX:1024
```

```
Files read from disk: 1495
Archive size: 4679509 bytes (4570 KiB)

Scan WARNINGS for files and folders:

/root/etc_backup/os-release : errno=2 : No such file or directory
/root/etc_backup/resolv.conf : errno=2 : No such file or directory
/root/etc_backup/mtab : errno=2 : No such file or directory
----------------
Scan WARNINGS: 3
root@eltex-practice2-pg2-v8:~# ls -lh /root/etc_backup.*
-rw-r--r-- 1 root root 4.5M Mar 13 09:11 /root/etc_backup.7z
-rw-r--r-- 1 root root 550K Mar 13 09:11 /root/etc_backup.tar.bz2
-rw-r--r-- 1 root root 584K Mar 13 09:11 /root/etc_backup.tar.gz
root@eltex-practice2-pg2-v8:~#
```

18. echo "$(date)" >> /etc/motd ; cat /etc/motd ; rsync -av /etc/ /root/etc_backup/ ; tar czf /root/etc_backup.tar.gz -C /root etc_backup ; ls -lh /root/etc_backup.tar.gz

```
root@eltex-practice2-pg2-v8:~# echo "$(date)" >> /etc/motd
root@eltex-practice2-pg2-v8:~# cat /etc/motd
Плотницкий Максим
Fri Mar 13 09:14:36 AM UTC 2026
root@eltex-practice2-pg2-v8:~#
```

```
root@eltex-practice2-pg2-v8:~# rsync -av /etc/ /root/etc_backup/
sending incremental file list
motd

sent 70,037 bytes  received 283 bytes  140,640.00 bytes/sec
total size is 2,366,832  speedup is 33.66
root@eltex-practice2-pg2-v8:~# tar czf /root/etc_backup.tar.gz -C /root etc_backup
root@eltex-practice2-pg2-v8:~# ls -lh /root/etc_backup.tar.gz
-rw-r--r-- 1 root root 584K Mar 13 09:15 /root/etc_backup.tar.gz
root@eltex-practice2-pg2-v8:~#
```

19. mkdir /tmp/compare_bz2 ; tar xjf /root/etc_backup.tar.bz2 -C /tmp/compare_bz2 ; diff -r /tmp/compare_bz2/etc_backup /root/etc_backup

```
root@eltex-practice2-pg2-v8:~# mkdir /tmp/compare_bz2
tar xjf /root/etc_backup.tar.bz2 -C /tmp/compare_bz2
mkdir: cannot create directory '/tmp/compare_bz2': File exists
root@eltex-practice2-pg2-v8:~# diff -r /tmp/compare_bz2/etc_backup /root/etc_backup
diff -r /tmp/compare_bz2/etc_backup/motd /root/etc_backup/motd
1a2
> Fri Mar 13 09:14:36 AM UTC 2026
diff: /tmp/compare_bz2/etc_backup/mtab: No such file or directory
diff: /root/etc_backup/mtab: No such file or directory
diff: /tmp/compare_bz2/etc_backup/os-release: No such file or directory
diff: /root/etc_backup/os-release: No such file or directory
diff: /tmp/compare_bz2/etc_backup/resolv.conf: No such file or directory
diff: /root/etc_backup/resolv.conf: No such file or directory
root@eltex-practice2-pg2-v8:~#
```

20. mkdir /root/etc_backup_gzip /root/etc_backup_7zip ; tar xzf /root/etc_backup.tar.gz -C /root/etc_backup_gzip ; 7z x /root/etc_backup.7z -o/root/etc_backup_7zip

```
root@eltex-practice2-pg2-v8:~# mkdir /root/etc_backup_gzip /root/etc_backup_7zip
root@eltex-practice2-pg2-v8:~# tar xzf /root/etc_backup.tar.gz -C /root/etc_backup_gzip
root@eltex-practice2-pg2-v8:~# 7z x /root/etc_backup.7z -o/root/etc_backup_7zip

7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
 64-bit locale=en_US.UTF-8 Threads:2 OPEN_MAX:1024

Scanning the drive for archives:
1 file, 4679509 bytes (4570 KiB)

Extracting archive: /root/etc_backup.7z
--
Path = /root/etc_backup.7z
Type = 7z
Physical Size = 4679509
Headers Size = 20847
Method = LZMA2:24 BCJ
Solid = +
Blocks = 2

Everything is Ok

Folders: 244
Files: 1508
Size:       38702832
Compressed: 4679509
root@eltex-practice2-pg2-v8:~# diff /root/etc_backup_gzip/etc_backup/motd /root/etc_backup_7zip/etc_
backup/motd
2d1
< Fri Mar 13 09:14:36 AM UTC 2026
root@eltex-practice2-pg2-v8:~#
```

Задания. Раздел 2.

Лог работы – команды и результаты в записать в файл practice2_${MYUSER}_part2.log, где MYUSER – переменная с именем пользователя практики 1.

1. Найдите все записи из лога загрузки, доступного через команду journalctl с опцией -b в первые 20 секунд с момента загрузки, начало загрузки определить автоматически.
2. Используя awk найдите все источники и их сообщения в файле auth.log (найдите его find), в названии источника удалите информацию об идентификаторе процесса при помощи sed, полученный результат отсортируйте по названию источника
3. Для результата из предыдущего пункта найдите количество повторений для каждого источника и выведите их в виде списка «число_повторений источник», результат отсортируйте по убыванию количества повторений
4. В файле /etc/passwd найдите всех пользователей в системе, у которых установлен shell /usr/sbin/nologin и выведите их в виде списка: «UID, username, список его групп», отсортированный в обратном порядке по UID, список групп используйте из файла /etc/group
5. Найдите в результате вывода dmesg все строки, содержащие слово 'kernel'
6. Подсчитайте количество строк в файле /var/log/kern.log
7. Отформатируйте вывод записей в /var/log/apt/history.log в следующем порядке, построчно: Commandline: ... ; Start-Date: ... ; End-Date: ...

1. **START=$(journalctl -b -o short-iso | head -1 | awk '{print $1}' ; END=$(date -d "$START +20 seconds" +"%Y-%m-%d %H:%M:%S") ;**
   journalctl -b --since="$START" --until="$END"

```
root@eltex-practice2-pg2-v8:~# START=$(journalctl -b -o short-iso | head -1 | awk '{print $1}')
root@eltex-practice2-pg2-v8:~# END=$(date -d "$START +20 seconds" +"%Y-%m-%d %H:%M:%S")
root@eltex-practice2-pg2-v8:~# journalctl -b --since="$START" --until="$END"
Mar 12 08:28:13 localhost kernel: Linux version 6.8.0-55-generic (buildd@lcy02-amd64-095) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-
Mar 12 08:28:13 localhost kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-55-generic root=/dev/mapper/ubuntu--vg-ubuntu--lv ro
Mar 12 08:28:13 localhost kernel: KERNEL supported cpus:
Mar 12 08:28:13 localhost kernel:   Intel GenuineIntel
Mar 12 08:28:13 localhost kernel:   AMD AuthenticAMD
Mar 12 08:28:13 localhost kernel:   Hygon HygonGenuine
Mar 12 08:28:13 localhost kernel:   Centaur CentaurHauls
Mar 12 08:28:13 localhost kernel:   zhaoxin   Shanghai
Mar 12 08:28:13 localhost kernel: BIOS-provided physical RAM map:
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000bffd9fff] usable
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x00000000bffda000-0x00000000bfffffff] reserved
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x00000000feffc000-0x00000000feffffff] reserved
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x0000000100000000-0x000000013fffffff] usable
Mar 12 08:28:13 localhost kernel: BIOS-e820: [mem 0x000000fd00000000-0x000000ffffffffff] reserved
Mar 12 08:28:13 localhost kernel: NX (Execute Disable) protection: active
Mar 12 08:28:13 localhost kernel: APIC: Static calls initialized
Mar 12 08:28:13 localhost kernel: SMBIOS 2.8 present.
Mar 12 08:28:13 localhost kernel: DMI: QEMU Standard PC (i440FX + PIIX, 1996), BIOS rel-1.17.0-0-gb52ca86e094d-prebuilt.qemu.org 0
Mar 12 08:28:13 localhost kernel: Hypervisor detected: KVM
Mar 12 08:28:13 localhost kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Mar 12 08:28:13 localhost kernel: kvm-clock: using sched offset of 1114298517595 cycles
Mar 12 08:28:13 localhost kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 8815905
Mar 12 08:28:13 localhost kernel: tsc: Detected 3400.000 MHz processor
```

2. AUTH_LOG=$(find /var/log -name auth.log -type f | head -1) ; awk '{process=$4; message=substr($0, index($0,$4)+length($4)+1); print process "|" message}' "$AUTH_LOG" | sed 's/\[[0-9]*\]//g' | sort -t'|' -k1,1

```
root@eltex-practice2-pg2-v8:~# AUTH_LOG=$(find /var/log -name auth.log -type f | head -1)
root@eltex-practice2-pg2-v8:~# awk '{process=$4; message=substr($0, index($0,$4)+length($4)+1); print process "|" message}' "$AUTH_LOG" | sed 's/\[[0-9]*\]//g' | sort -t'|' -k1,1
Accepted|password for root from 172.16.8.4 port 52968 ssh2
Accepted|password for root from 172.16.8.4 port 55466 ssh2
Accepted|password for root from 172.16.8.4 port 56442 ssh2
Accepted|publickey for root from 172.16.8.4 port 35494 ssh2: RSA SHA256:8lhcXdX3ramGxwOtpvRHtVOTYS4q/Z0wLditSdu9iAU
Accepted|publickey for root from 172.16.8.4 port 38330 ssh2: RSA SHA256:8lhcXdX3ramGxwOtpvRHtVOTYS4q/Z0wLditSdu9iAU
Accepted|publickey for root from 172.16.8.4 port 46836 ssh2: RSA SHA256:8lhcXdX3ramGxwOtpvRHtVOTYS4q/Z0wLditSdu9iAU
Accepted|publickey for root from 172.16.8.4 port 55582 ssh2: RSA SHA256:8lhcXdX3ramGxwOtpvRHtVOTYS4q/Z0wLditSdu9iAU
```

3. awk '{process=$4; message=substr($0, index($0,$4)+length($4)+1); print process "|" message}' "$AUTH_LOG" | sed 's/\[[0-9]*\]//g' | cut -d'|' -f1 | sort | uniq -c | sort -rn | awk '{print $1, $2}'

```
root@eltex-practice2-pg2-v8:~# awk '{process=$4; message=substr($0, index($0,$4)+length($4)+1); print process "|" message}' "$AUTH_LOG" | sed 's/\[[0-9]*\]//g' |
386 pam_unix(cron:session):
52 pam_unix(su-l:session):
32 Loading
28 (to
18 add
```

4.

```
root@eltex-practice2-pg2-v8:~# grep "/usr/sbin/nologin" /etc/passwd | cut -d: -f1,3 | while IFS=: read user uid; do
    groups=$(groups "$user" 2>/dev/null | cut -d: -f2- | sed 's/^ //' | tr ' ' ',')
    echo "$uid,$user,$groups"
done | sort -nr -t, -k1
65534,nobody,nogroup
1003,user3,user3,users,users_group
998,systemd-network,systemd-network
997,systemd-timesync,systemd-timesync
992,systemd-resolve,systemd-resolve
991,polkitd,polkitd
989,fwupd-refresh,fwupd-refresh
109,sshd,nogroup
108,usbmux,plugdev
107,landscape,landscape
105,tcpdump,tcpdump
104,uuidd,uuidd
103,syslog,syslog,adm
101,messagebus,messagebus
42,_apt,nogroup
39,irc,irc
38,list,list
34,backup,backup
33,www-data,www-data
13,proxy,proxy
10,uucp,uucp
9,news,news
8,mail,mail
7,lp,lp
6,man,man
5,games,games
3,sys,sys
2,bin,bin
1,daemon,daemon
```

5. grep "/usr/sbin/nologin" /etc/passwd | cut -d: -f1,3 | while IFS=: read user uid; do
groups=$(groups "$user" 2>/dev/null | cut -d: -f2- | sed 's/^ //' | tr ' ' ',')
echo "$uid,$user,$groups"
done | sort -nr -t, -k1

```
root@eltex-practice2-pg2-v8:~# dmesg | grep -i kernel
[    0.000000] KERNEL supported cpus:
[    0.154783] Booting paravirtualized kernel on KVM
[    0.158641] Kernel command line: BOOT_IMAGE=/vmlinuz-6.8.0-55-generic root=/dev/mapper/ubuntu--vg-ubuntu--lv ro
[    0.158798] Unknown kernel command line parameters "BOOT_IMAGE=/vmlinuz-6.8.0-55-generic", will be passed to user space.
[    0.410943] Memory: 3933500K/4193760K available (22528K kernel code, 4443K rwdata, 14344K rodata, 4988K init, 4716K bss, 260000K reserved, 0K cma-reserved)
[    0.584070] DMA: preallocated 512 KiB GFP_KERNEL pool for atomic allocations
[    0.584125] DMA: preallocated 512 KiB GFP_KERNEL|GFP_DMA pool for atomic allocations
[    0.584177] DMA: preallocated 512 KiB GFP_KERNEL|GFP_DMA32 pool for atomic allocations
[    1.241948] Loaded X.509 cert 'Build time autogenerated kernel key: e0bedf6fa049a1e09bd65ebab3904f14cdd5bc68'
[    1.254738] Loaded X.509 cert 'Canonical Ltd. Kernel Module Signing: 88f752e560a1e0737e31163a466ad7b70a850c19'
[    1.393339] Loaded X.509 cert 'Build time autogenerated kernel key: e0bedf6fa049a1e09bd65ebab3904f14cdd5bc68'
[    1.430369] Freeing unused kernel image (initmem) memory: 4988K
[    1.431472] Write protecting the kernel read-only data: 38912k
[    1.433109] Freeing unused kernel image (rodata/data gap) memory: 2040K
[    9.570285] systemd[1]: Listening on systemd-udevd-kernel.socket - udev Kernel Socket.
[    9.611624] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File System...
[    9.619118] systemd[1]: Mounting sys-kernel-tracing.mount - Kernel Trace File System...
[    9.775067] systemd[1]: Starting modprobe@configfs.service - Load Kernel Module configfs...
[    9.853327] systemd[1]: Starting modprobe@dm_mod.service - Load Kernel Module dm_mod...
[    9.929033] systemd[1]: Starting modprobe@drm.service - Load Kernel Module drm...
[   10.081615] systemd[1]: Starting modprobe@efi_pstore.service - Load Kernel Module efi_pstore...
[   10.127197] systemd[1]: Starting modprobe@fuse.service - Load Kernel Module fuse...
[   10.170297] systemd[1]: Starting modprobe@loop.service - Load Kernel Module loop...
[   10.399269] systemd[1]: Starting systemd-modules-load.service - Load Kernel Modules...
[   10.412583] systemd[1]: Starting systemd-remount-fs.service - Remount Root and Kernel File Systems...
root@eltex-practice2-pg2-v8:~#
```

6. wc -l /var/log/kern.log

```
root@eltex-practice2-pg2-v8:~# wc -l /var/log/kern.log
2884 /var/log/kern.log
```

7. awk '/^Start-Date:/ {start=$0} /^Commandline:/ {cmd=$0} /^End-Date:/ {end=$0; print cmd " ; " start " ; " end}' /var/log/apt/history.log

```
root@eltex-practice2-pg2-v8:~# awk '/^Start-Date:/ {start=$0} /^Commandline:/ {cmd=$0} /^End-Date:/ {end=$0; print cmd " ; " start " ; " end}' /var/log/apt/histo
y.log
Commandline: apt-get install --reinstall apt ; Start-Date: 2025-02-16  20:51:22 ; End-Date: 2025-02-16  20:51:24
Commandline: apt-get install --reinstall base-files ; Start-Date: 2025-02-16  20:51:26 ; End-Date: 2025-02-16  20:51:28
Commandline: apt-get install --reinstall base-passwd ; Start-Date: 2025-02-16  20:51:31 ; End-Date: 2025-02-16  20:51:32
Commandline: apt-get install --reinstall bash ; Start-Date: 2025-02-16  20:51:35 ; End-Date: 2025-02-16  20:51:37
Commandline: apt-get install --reinstall bsdutils ; Start-Date: 2025-02-16  20:51:39 ; End-Date: 2025-02-16  20:51:41
Commandline: apt-get install --reinstall coreutils ; Start-Date: 2025-02-16  20:51:44 ; End-Date: 2025-02-16  20:51:45
Commandline: apt-get install --reinstall dash ; Start-Date: 2025-02-16  20:51:48 ; End-Date: 2025-02-16  20:51:49
Commandline: apt-get install --reinstall debconf ; Start-Date: 2025-02-16  20:51:52 ; End-Date: 2025-02-16  20:51:53
Commandline: apt-get install --reinstall debianutils ; Start-Date: 2025-02-16  20:51:56 ; End-Date: 2025-02-16  20:51:58
Commandline: apt-get install --reinstall diffutils ; Start-Date: 2025-02-16  20:52:00 ; End-Date: 2025-02-16  20:52:02
Commandline: apt-get install --reinstall dpkg ; Start-Date: 2025-02-16  20:52:04 ; End-Date: 2025-02-16  20:52:06
```

Задания. Раздел 3.

Лог работы – команды и результаты в записать в файл practice2_${MYUSER}_part3.log, где MYUSER – переменная с именем пользователя практики 1.

1. Зарегистрируйтесь на сайте github.com, если еще не сделали этого ранее
2. Установите параметры git в соответствии с вашими учетными данными:
   git config --global user.name "имя пользователя github"
   git config --global user.email "ваш e-mail, использованный при регистрации"
   git config --global init.defaultBranch main
3. Установите непосредственно программу **git** и набор утилит для управления github из командной строки **gh**:
   apt update
   apt install git gh
4. Создайте ключ ssh:
   ssh-keygen -t ed25519 -f ~/.ssh/название_ключа (имя пользователя github или имя пользователя для первой практики, или на ваш вкус)
5. Подключите этот комьютер к github:
   gh auth login
   ? What account do you want to log into? **GitHub.com**
   ? What is your preferred protocol for Git operations on this host? **SSH**
   ? Upload your SSH public key to your GitHub account? **/root/.ssh/название_ключа.pub**
   ? Title for your SSH key: eltex-practice2-pg**X**-v**Y**
   ? How would you like to authenticate GitHub CLI? **Login with a web browser**

   Если вы выполняете команды из текстовой консоли, то браузер не откроется автоматически, необходимо на вашем рабочем компьютере открыть сайт https://github.com/login/device и ввести код авторизации из лога: «First copy your one-time code: **...**»

2. git config --global user.name "particsh" ; git config --global user.email "maxim.plotnitskiyreborn@gmail.com" ; git config --global init.defaultBranch main

```
root@eltex-practice2-pg2-v8:~# git config --global user.name "particsh"
root@eltex-practice2-pg2-v8:~# git config --global user.email "maxim.plotnitskiyreborn@gmail.com"
root@eltex-practice2-pg2-v8:~# git config --global init.defaultBranch main
root@eltex-practice2-pg2-v8:~# apt update
```

3. apt update ; apt install git gh

```
root@eltex-practice2-pg2-v8:~# apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://ru.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://ru.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://ru.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:6 http://ru.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,808 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.6 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.8 k
Get:11 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:12 http://ru.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:13 http://ru.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:14 http://ru.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,566 kB]
Get:15 http://ru.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:16 http://ru.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:17 http://ru.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,288 B]
Get:18 http://ru.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B
Get:19 http://ru.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [32.0 kB]
Get:20 http://ru.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.3 kB]
Get:21 http://ru.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB
Get:22 http://ru.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:23 http://ru.archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:24 http://ru.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B
Fetched 4,515 kB in 1s (3,895 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
87 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
root@eltex-practice2-pg2-v8:~# apt install git gh
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
The following NEW packages will be installed:
  gh
0 upgraded, 1 newly installed, 0 to remove and 87 not upgraded.
Need to get 8,836 kB of archives.
After this operation, 45.4 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://ru.archive.ubuntu.com/ubuntu noble-updates/universe amd64 gh amd64 2.45.0-1ubuntu0.3 [8,836 kB]
Fetched 8,836 kB in 1s (8,565 kB/s)
Selecting previously unselected package gh.
(Reading database ... 125245 files and directories currently installed.)
Preparing to unpack .../gh_2.45.0-1ubuntu0.3_amd64.deb ...
Unpacking gh (2.45.0-1ubuntu0.3) ...
Setting up gh (2.45.0-1ubuntu0.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-55-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-101-generic.
```

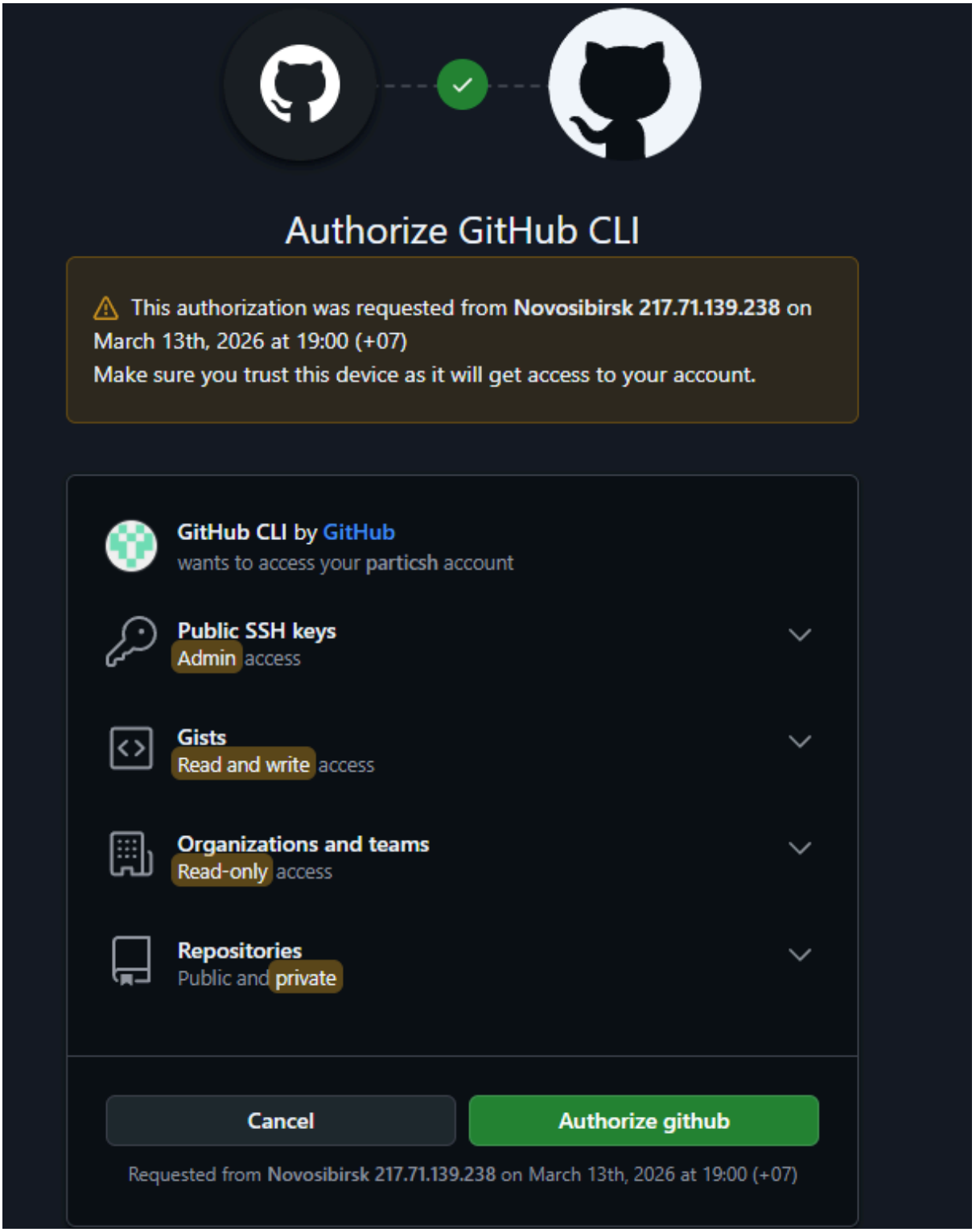
4. **ssh-keygen -t ed25519 -f ~/.ssh/particsh**

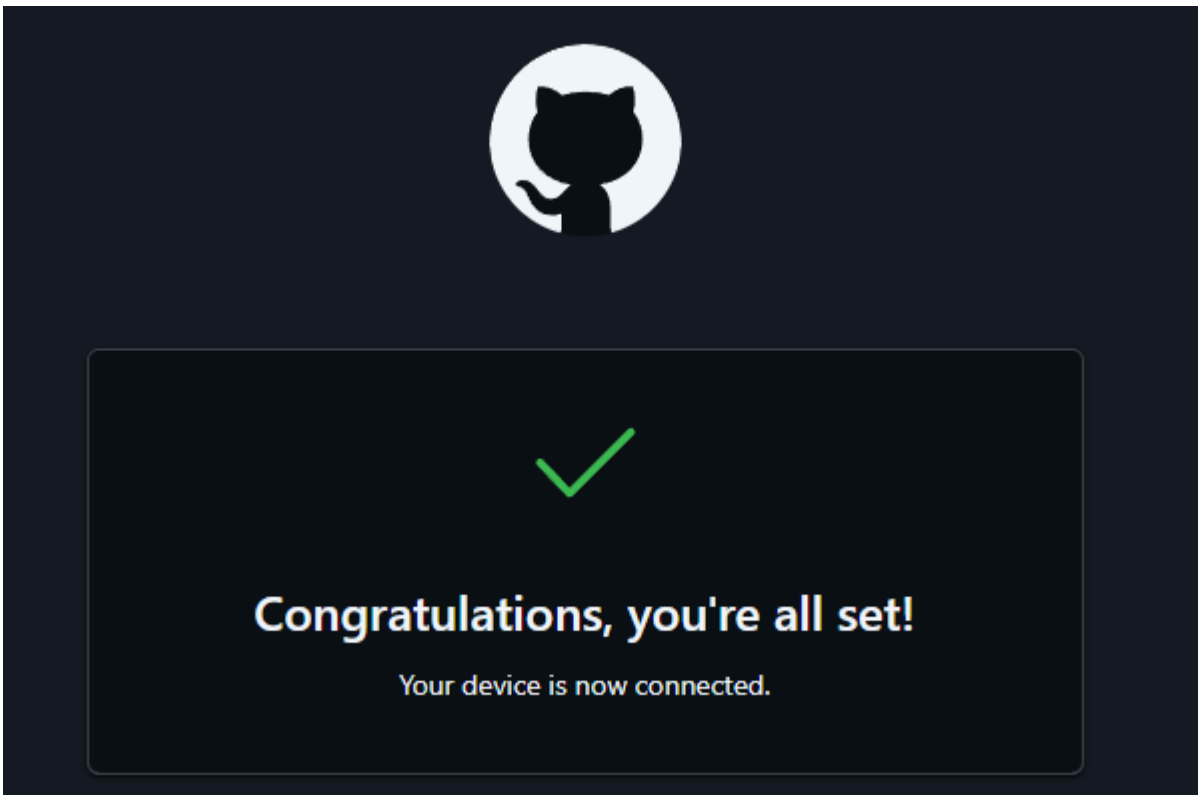
```
root@eltex-practice2-pg2-v8:~# ssh-keygen -t ed25519 -f ~/.ssh/particsh
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/particsh
Your public key has been saved in /root/.ssh/particsh.pub
The key fingerprint is:
SHA256:8VKdJXKXG2IRpqh80oXSPnMqDvpHAfz3tGAFzh9UCjc root@eltex-practice2-pg2-v8
The key's randomart image is:
+--[ED25519 256]--+
|    .    o.E.Boo. |
|     o  + *.Oo=o  |
|      o. B.*.o. o |
|      .o=+*..   .  |
|       ++S++.      |
|       .o *o       |
|     ... .         |
|    . o..          |
|   .....           |
+----[SHA256]-----+
root@eltex-practice2-pg2-v8:~#
```

5. gh auth login

```
root@eltex-practice2-pg2-v8:~# gh auth login
? What account do you want to log into? GitHub.com
? What is your preferred protocol for Git operations on this host? SSH
? Upload your SSH public key to your GitHub account? /root/.ssh/particsh.pub
? Title for your SSH key: eltex-practice2-pg2-v8
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: AA67-7EF3
Press Enter to open github.com in your browser...
! Failed opening a web browser at https://github.com/login/device
  exec: "xdg-open,x-www-browser,www-browser,wslview": executable file not found in $PATH
  Please try entering the URL in your browser manually
```


Authorize GitHub CLI
This authorization was requested from Novosibirsk 217.71.139.238 on March 13th, 2026 at 19:00 (+07)
Make sure you trust this device as it will get access to your account.
GitHub CLI by GitHub
wants to access your particsh account
Public SSH keys
Admin access
Gists
Read and write access
Organizations and teams
Read-only access
Repositories
Public and private
Cancel
Authorize github
Requested from Novosibirsk 217.71.139.238 on March 13th, 2026 at 19:00 (+07)




```
✓ Authentication complete.
- gh config set -h github.com git_protocol ssh
✓ Configured git protocol
! Authentication credentials saved in plain text
✓ Uploaded the SSH key to your GitHub account: /root/.ssh/particsh.pub
✓ Logged in as particsh
```

Задания. Раздел 3.

6. Укажите автоматическую авторизацию только для сайта github.com:
   gh auth setup-git --hostname github.com
7. Убедиться, что все в порядке можно командой: gh auth status
8. Создайте репозиторий на сервере github:
   gh repo create --public название_репозитория
9. Пора наполнить его:
   mkdir ~/название_репозитория && cd ~/название_репозитория
   echo "# краткое_описание_репозитория" >> README.md
   git init
   git add README.md
   git commit -m "first commit"
   git branch -M main
   git remote add origin https://github.com/имя_пользователя_github/название_репозитория.git
   git status
   git push -u origin main
   mkdir -p module1/task1
   mkdir module1/task2
   mkdir module1/task3
   (скопируйте с сервера для 1 практики отчеты в каталоги task* используя команду scp)
   git add .
   git status
   git commit (введите комментарий)
   git push

6 - 7. gh auth setup-git --hostname github.com ; gh auth status

```
root@eltex-practice2-pg2-v8:~# gh auth setup-git --hostname github.com
root@eltex-practice2-pg2-v8:~# gh auth status
github.com
  ✓ Logged in to github.com account particsh (/root/.config/gh/hosts.yml)
  - Active account: true
  - Git operations protocol: ssh
  - Token: gho_************************************
  - Token scopes: 'admin:public_key', 'gist', 'read:org', 'repo'
root@eltex-practice2-pg2-v8:~#
```

8. gh repo create --public eltex_module1

```
root@eltex-practice2-pg2-v8:~# gh repo create --public eltex_module1
✓ Created repository particsh/eltex_module1 on GitHub
  https://github.com/particsh/eltex_module1
```

9. mkdir ~/eltex_module1 && cd ~/eltex_module1 ; echo "# eltex module1" >> [README.md](https://github.com/particsh/eltex_module1/blob/main/README.md) ; git add [README.md](https://github.com/particsh/eltex_module1/blob/main/README.md) ; git commit -m "first commit"; git branch -M main ; git remote add origin https://github.com/particsh/eltex_module1.git ; git status ; git push -u origin main ; scp -P 44556 '.\otchet_1 (1).pdf' eltex-pg2-v8@217.71.138.1:~ ; scp otchet_1.pdf root@172.16.9.208:~ ; scp eltex-pg2-v8_part.log eltex-pg2-v8_part3.log eltex-pg2-v8_part3_gt50M.log root@172.16.9.208:~ ; cp ../eltex-pg2-v8_part.log

../eltex-pg2-v8_part3.log ../eltex-pg2-v8_part3_gt50M.log ./module1/task1 ; git add * ; git status ; git commit -m "task1" ;

```
root@eltex-practice2-pg2-v8:~# gh repo create --public eltex_module1
✓ Created repository particsh/eltex_module1 on GitHub
  https://github.com/particsh/eltex_module1
root@eltex-practice2-pg2-v8:~# mkdir ~/eltex_module1 && cd ~/eltex_module1'
> ^C
root@eltex-practice2-pg2-v8:~# mkdir ~/eltex_module1 && cd ~/eltex_module1
root@eltex-practice2-pg2-v8:~/eltex_module1# git init
Initialized empty Git repository in /root/eltex_module1/.git/
root@eltex-practice2-pg2-v8:~/eltex_module1# git add README.md
fatal: pathspec 'README.md' did not match any files
root@eltex-practice2-pg2-v8:~/eltex_module1# git init
Reinitialized existing Git repository in /root/eltex_module1/.git/
root@eltex-practice2-pg2-v8:~/eltex_module1# echo "# eltex module1" >> README.md
root@eltex-practice2-pg2-v8:~/eltex_module1# git add README.md
root@eltex-practice2-pg2-v8:~/eltex_module1# git commit -m "first commit"
[main (root-commit) 59e21f6] first commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
root@eltex-practice2-pg2-v8:~/eltex_module1# git branch -M main
root@eltex-practice2-pg2-v8:~/eltex_module1# git remote add origin https://github.com/particsh/eltex_module1.git
root@eltex-practice2-pg2-v8:~/eltex_module1# git status
On branch main
nothing to commit, working tree clean
root@eltex-practice2-pg2-v8:~/eltex_module1# git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 237 bytes | 237.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/particsh/eltex_module1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
root@eltex-practice2-pg2-v8:~/eltex_module1# mkdir -p module1/task1
root@eltex-practice2-pg2-v8:~/eltex_module1# mkdir -p module1/task2
root@eltex-practice2-pg2-v8:~/eltex_module1# mkdir -p module1/task3
root@eltex-practice2-pg2-v8:~/eltex_module1#
```

```
PS C:\Users\particsh\Downloads> scp -P 44556 '.\otchet_1 (1).pdf' eltex-pg2-v8@217.71.138.1:~
The authenticity of host '[217.71.138.1]:44556 ([217.71.138.1]:44556)' can't be established.
ED25519 key fingerprint is SHA256:9EeAnPIp/wDj41XS55iG0Eff5a90ZLW9TTAI37NiBb0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '[217.71.138.1]:44556' (ED25519) to the list of known hosts.
eltex-pg2-v8@217.71.138.1's password:
otchet_1 (1).pdf                                              100%  986KB   2.7MB/s   00:00
PS C:\Users\particsh\Downloads>
```

```
eltex-pg2-v8@eltex-2026-spring:~$ scp otchet_1.pdf root@172.16.9.208:~
otchet_1.pdf                                  100%  986KB 144.1MB/s   00:00
```

```
eltex-pg2-v8@eltex-2026-spring:~$ scp eltex-pg2-v8_part.log eltex-pg2-v8_part3.log eltex-pg2-v8_part3_gt50M.log root@172.16.9.208:~
eltex-pg2-v8_part.log                                                                                    100%  47
eltex-pg2-v8_part3.log                                                                                   100%  46
eltex-pg2-v8_part3_gt50M.log                                                                             100%  53
```

```
root@eltex-practice2-pg2-v8:~/eltex_module1# cp ../eltex-pg2-v8_part.log ../eltex-pg2-v8_part3.log ../eltex-pg2
```

```
root@eltex-practice2-pg2-v8:~/eltex_module1# git add *
root@eltex-practice2-pg2-v8:~/eltex_module1# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   module1/task1/eltex-pg2-v8_part.log
        new file:   module1/task1/eltex-pg2-v8_part3.log
        new file:   module1/task1/eltex-pg2-v8_part3_gt50M.log
        new file:   module1/task1/otchet_1.pdf

root@eltex-practice2-pg2-v8:~/eltex_module1# git commit -m "task1"
[main d2a3dda] task1
 4 files changed, 60 insertions(+)
 create mode 100644 module1/task1/eltex-pg2-v8_part.log
 create mode 100644 module1/task1/eltex-pg2-v8_part3.log
 create mode 100644 module1/task1/eltex-pg2-v8_part3_gt50M.log
 create mode 100644 module1/task1/otchet_1.pdf
root@eltex-practice2-pg2-v8:~/eltex_module1# git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 877.54 KiB | 31.34 MiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/particsh/eltex_module1.git
   59e21f6..d2a3dda  main -> main
root@eltex-practice2-pg2-v8:~/eltex_module1#
```

Задания. Раздел 3.

10. Клонируйте ваш репозиторий:
    mkdir ~/название_репозитория_clone && cd ~/название_репозитория_clone
    git clone https://github.com/**имя_пользователя_github/название_репозитория**.git . (nb! точка)
11. Внесите изменение в основном каталоге, не забыв про git add/commit/push, выполните в каталоге с клоном:
    - git fetch
    - git log origin/main
    - git diff main origin/main
    - git merge (для автоматического слияния используйте git pull вместо fetch+merge)
12. Повторите п.9 начиная с создания каталогов для отчета по второй практике
13. Отправьте ссылку на ваш репозиторий преподавателю в telegram, если в имени нет ФИО – дополните сообщение, пожалуйста. Вы конечно представлялись в общем чате, но искать в истории немножко тоскливо.

10. mkdir ~/eltex_module1_clone && cd eltex_module1_clone/ ; git clone https://github.com/particsh/eltex_module1.git .

```
root@eltex-practice2-pg2-v8:~# mkdir eltex_module1_clone && cd ~/eltex_module1_clone
root@eltex-practice2-pg2-v8:~/eltex_module1_clone#
```

```
root@eltex-practice2-pg2-v8:~/eltex_module1_clone# git clone https://github.com/particsh/eltex_module1.git
Cloning into 'eltex_module1'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 11 (delta 0), reused 11 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), 877.74 KiB | 1.57 MiB/s, done.
```

11. echo "Hallo again" >> README.md ; git add * ; git commit -m "For task 11" ; git push ; cd .. ; cd eltex_module1_clone/ ; git fetch ; git log origin/main ; git diff main origin/main ; git merge

```
root@eltex-practice2-pg2-v8:~/eltex_module1_clone/eltex_module1# git add README.md
root@eltex-practice2-pg2-v8:~/eltex_module1_clone/eltex_module1# git commit -m "second commit"
[main fa2794d] second commit
 1 file changed, 1 insertion(+)
root@eltex-practice2-pg2-v8:~/eltex_module1_clone/eltex_module1# git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 304 bytes | 304.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/particsh/eltex_module1.git
   d2a3dda..fa2794d  main -> main
root@eltex-practice2-pg2-v8:~/eltex_module1_clone/eltex_module1# git fetch
root@eltex-practice2-pg2-v8:~/eltex_module1_clone/eltex_module1# git log origin/main
commit fa2794db4734bdbfb06fa1ffe313fdad1336616a (HEAD -> main, origin/main, origin/HEAD)
Author: particsh <maxim.plotnitskiyreborn@gmail.com>
Date:   Fri Mar 13 12:34:11 2026 +0000

    second commit

commit d2a3ddae25c6aa81198d73d9ec99e820aaae5d71
Author: particsh <maxim.plotnitskiyreborn@gmail.com>
Date:   Fri Mar 13 12:26:17 2026 +0000

    task1

commit 59e21f659043571d42645777001810b6d463d969
Author: particsh <maxim.plotnitskiyreborn@gmail.com>
Date:   Fri Mar 13 12:08:40 2026 +0000

    first commit
root@eltex-practice2-pg2-v8:~/eltex_module1_clone/eltex_module1# git diff main origin/main
root@eltex-practice2-pg2-v8:~/eltex_module1_clone/eltex_module1# git merge
Already up to date.
```